



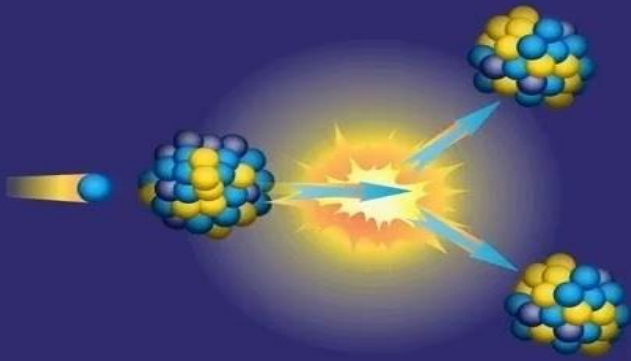
NUCLEAR POWER STATION

INTRODUCTION:

- ▶ Nuclear power station, nuclear energy is converted into electrical energy .The main atomic fuels used are Uranium and Thorium. These are subjected to nuclear fission in a special apparatus known as reactor. The heat energy thus released is utilized in raising steam at high temperature and pressure. The steam runs the steam turbine which converts steam energy into mechanical energy. The turbine drives the generator which converts mechanical energy into electrical energy.

NUCLEAR FISSION:

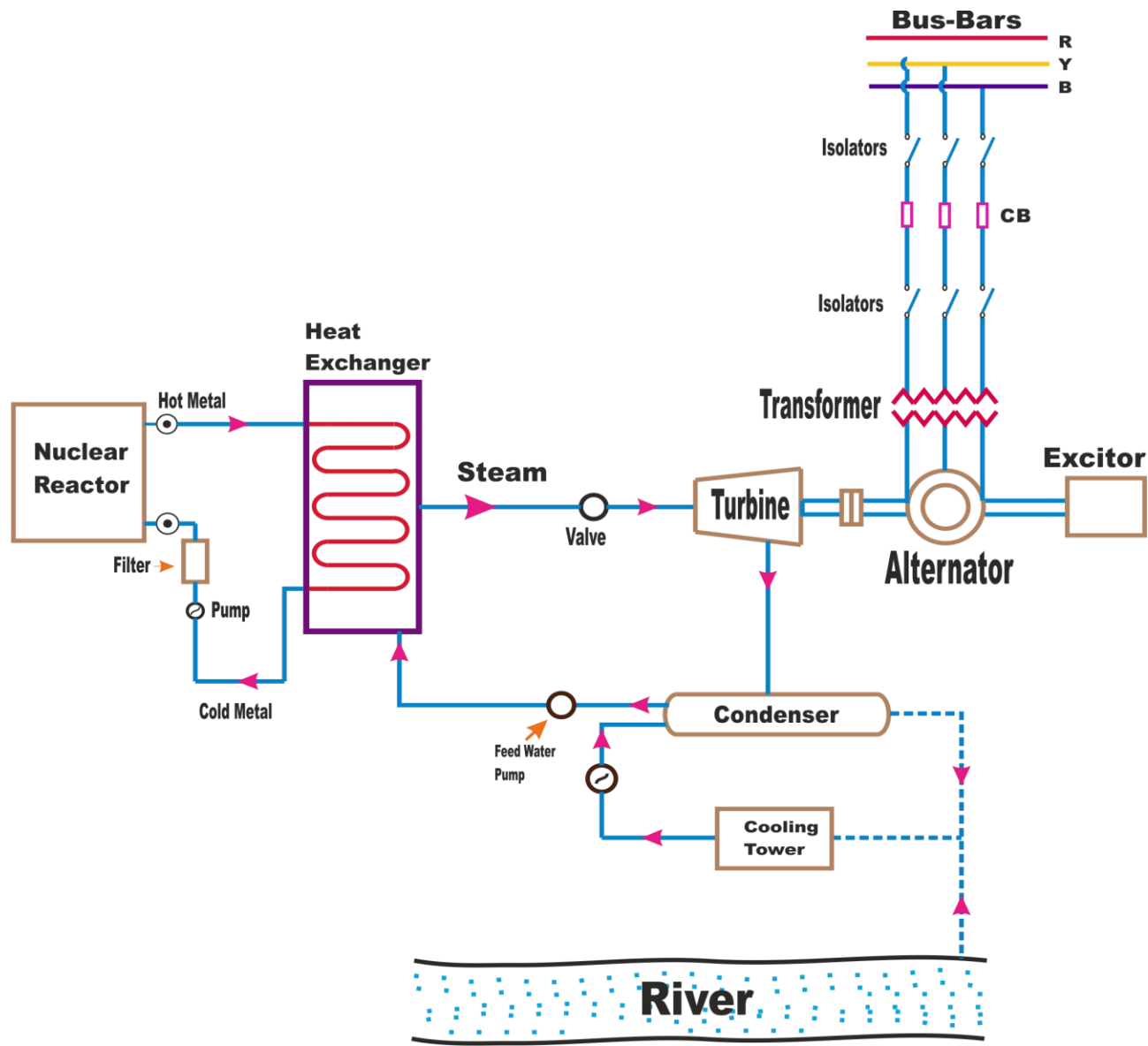
Fission



Splits a larger atom into
2 or more smaller ones

In nuclear fission, atoms are split apart, which releases energy. All nuclear power plants use nuclear fission, and most nuclear power plants use uranium atoms. During nuclear fission, a neutron collides with a uranium atom and splits it, releasing a large amount of energy in the form of heat and radiation. More neutrons are also released when a uranium atom splits. These neutrons continue to collide with other uranium atoms, and the process repeats itself over and over again. This process is called a nuclear chain reaction. This reaction is controlled in nuclear power plant reactors to produce a desired amount of heat.

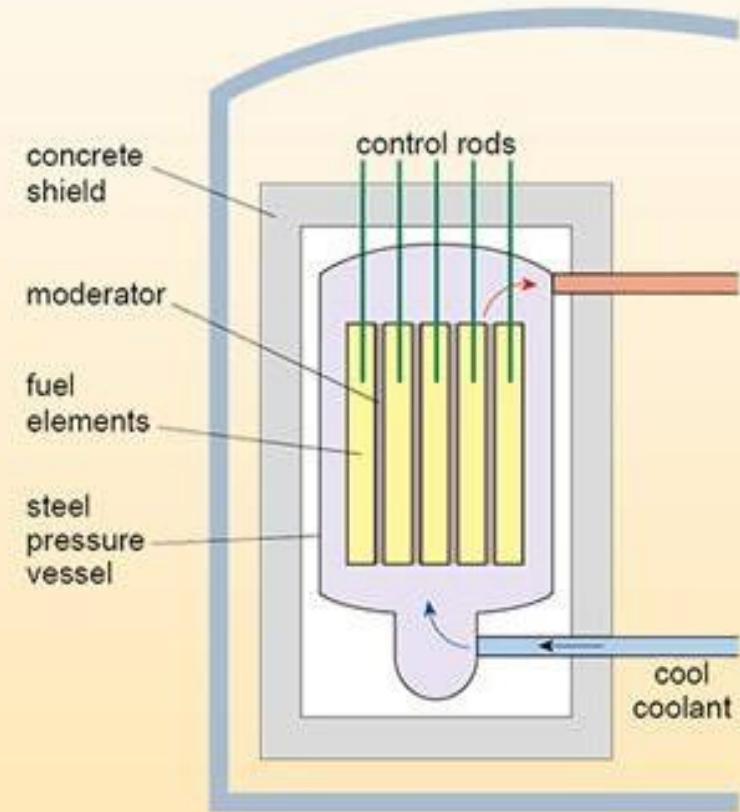
- * Uranium235(isotope of uranium) is used as nuclear fuel.
- * Nuclear fusion takes place in sun and other stars.



BLOCK SCHEMATIC FOR NUCLEAR POWER PLANT

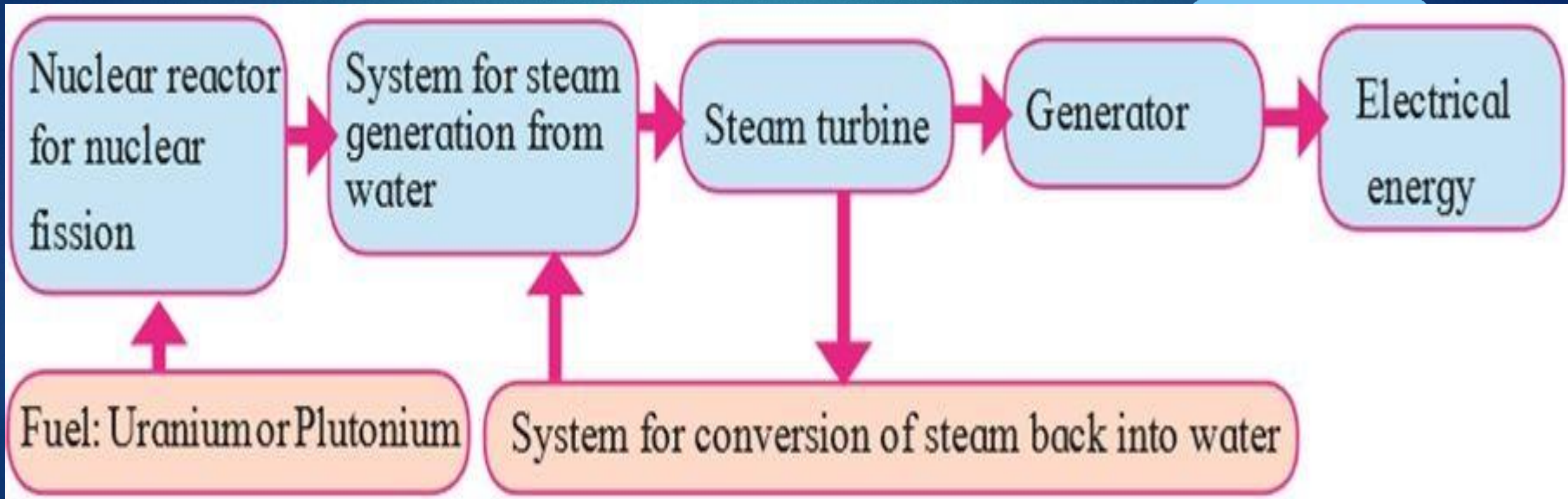
COMPONENTS:

- **Nuclear Reactor** is an apparatus in which nuclear fuel is subjected to nuclear fission which controls the chain reaction.
- It houses fuel rods of uranium, moderator (graphite) and control rods.
- **Fuel rods** consists of the fission material and releases large amount of energy when bombarded with slow moving neutrons.
- **Moderator** encloses the fuel rods which slows down the neutrons before they bombards the fuel rods
- **Control rods** absorb neutrons and regulate the supply of neutrons.
By adjusting the position of control rods, intensity of chain reaction can be increased or decreased.
- Heat produced in the reactor is removed by a **coolant**, generally sodium metal.
- **Heat Exchanger** Coolant gives up the heat to the heat exchanger which is utilized in raising the steam.
- **Steam Turbine** Steam produced in the heat exchanger is led to the steam turbine .After doing work, steam is exhausted to condenser. Condenser condenses the steam



*In heavy water reactors deuterium is used as the coolant.

WORKING OF POWER PLANT:



ADVANTAGES:

- ▶ 1. The amount of fuel required is quite small, therefore, there is no problem of transportation, storage etc.
- ▶ 2. These plants need less area as compared to any other type.
- ▶ 3. These plants can be located near the load centers, therefore, primary distribution cost is reduced.
- ▶ 4. These plants are most economical in large capacity.
- ▶ 5. There are large deposits of nuclear fuels available all over the world. Therefore such plants can ensure continued supply of electrical energy for thousands of years.

DISADVANTAGES:

- ▶ 1. The initial capital cost is very high as compared to other types of power plants.
- ▶ 2. The erection and commissioning of the plant requires greater knowledge.
- ▶ 3. The fission by products are generally radio-active and may cause amount of radio-active pollution.
- ▶ 4. The fuel used is expensive a dangerous
- ▶ 5. The disposal of the products, which are radio-active, is a big problem.

THANK YOU

